

DATE FORMAT CONVERSION

PROBLEM STATEMENT:

A computer program is required to determine whether or not a date that is provided as input, occurred during a leap year. In addition to outputting this determination for each date input, some summary outputs are also to be provided. This program will be incorporated in a larger program that converts date formats from MDY to Ordinal Dates.

All input will come from a keyboard and all output will be sent to a screen.

The initial input will be the client number. Because the program must accommodate an unknown number of dates, the prompt and response method will be used to determine when all the dates have been input. Accordingly, the detail inputs will be a value for the response variable (Y or N), and the date in month/day/year format.

After each date is entered, the program will immediately output the year and whether or not that year is leap year. This output will take the form of the word TRUE or FALSE.

Following the last date to be input, a value of N for the response variable will signal that input is complete and that summary values should be output now. The summary outputs are the client number, a count of how many input dates were leap years, a count of how many input dates were not leap years, the percentage of dates entered that were leap years, and finally the year of the most recent leap year in this data set.

In the event of multiple instances of the latest, only the first instance is to be reported.

INPUT/OUTPUT DESIGN:

INPUT:

client_nr: the number assigned to client requesting this report

more: the response variable a value of Y indicates there are addition dates to process, N indicates end of input

mdy_date: the date to be converted in month/day/year format

OUTPUT:

year: the four digit year of the date that was input

leap_year: the word TRUE or FALSE corresponding to whether or not the year was a leap year

client_nr: the number assigned to client requesting this report

count_ly: a count of how many of the input dates occurred during a leap year

count_not_ly: a count of how many of the input dates did not occur during a leap year

percent_ly: the percentage of dates entered that were leap years, and finally

most_recent_ly: the year of the most recent leap year in this data set